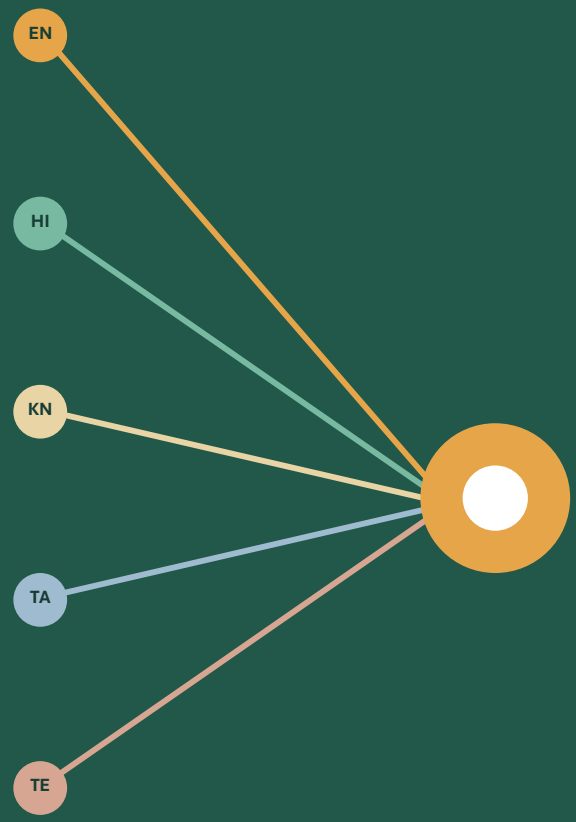
UNIVERSITY PROJECT / PRJ_277

360°
News coverage research system

A multilingual workflow to collect, analyze, compare and trace public reporting on Government of India initiatives.

FUNCTIONAL 60% MILESTONE

Evidence-linked · Politically neutral · Local-first



ONE STORY SET

METHOD • PRODUCT • LIMITS

A working research slice, measured by capability

100
synthetic articles

5
supported languages

10
demo story groups

6 / 10
capability groups in scope

20 DEMO ARTICLES PER LANGUAGE



Read this correctly

The 60% figure tracks six working capability groups out of ten. It is a project-scope measure—not a grade, code-completion percentage, or claim of production readiness.

SYNTHETIC DEMO DATA

One public initiative can become five different samples

What a researcher must connect

01 Discovery

Which reports are in scope?

02 Comparability

Are language and outlet samples comparable?

03 Interpretation

What is reported, and what is inferred?

04 Traceability

Can each comparison be checked against its sources?

STUDY QUESTION

How does coverage differ across languages and news sources?

The app reports observed sample differences. It does not infer truth, intent, political bias, or outlet quality.

DESCRIPTIVE COMPARISON

Start with the question—not an unbounded crawl

A researcher defines the topic, aliases, time window, languages and optional outlet sample.

NEW COVERAGE SEARCH

Topic or initiative

Swachh Bharat Mission

Alternate names

Clean India campaign · SBM

PUBLICATION DATES

FROM 01 AUG 2026

THROUGH 31 AUG 2026

LANGUAGE SAMPLE

EN HI KN TA TE

A bounded online sample

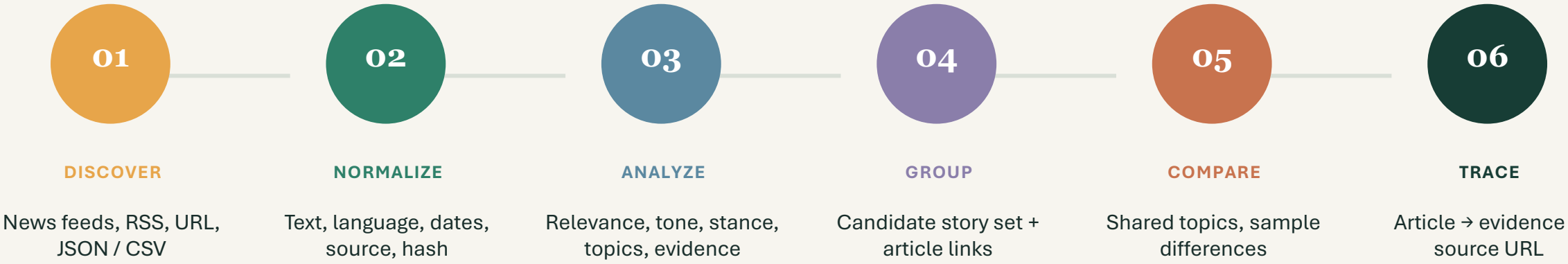
- **Google News RSS + Bing News RSS**
- Selected dates and languages
- Optional source-domain filter
- Small per-language result cap

Not exhaustive

Search providers and publisher access limit recall.

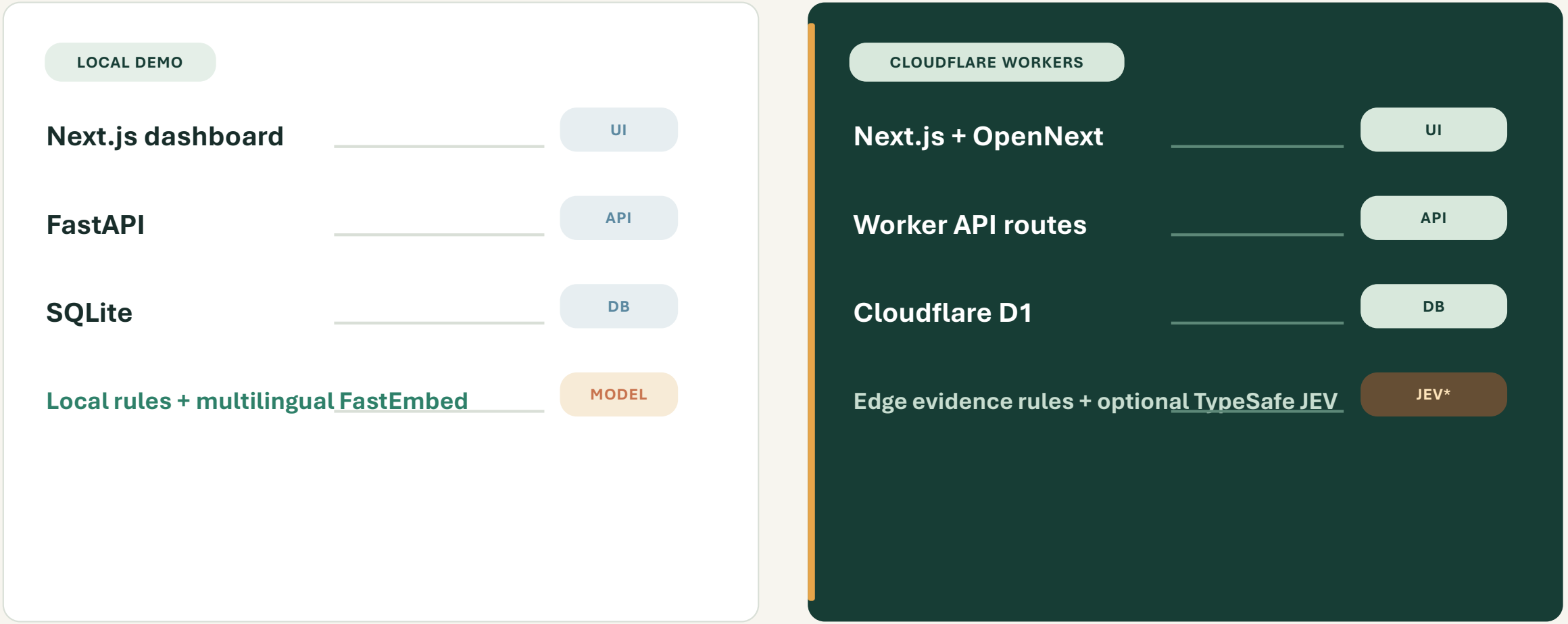
From search prompt to inspectable comparison

Every processing stage writes records that downstream screens can trace.



Human review stays in the loop Low support stays uncertain; comparison statements link to the article IDs behind them.

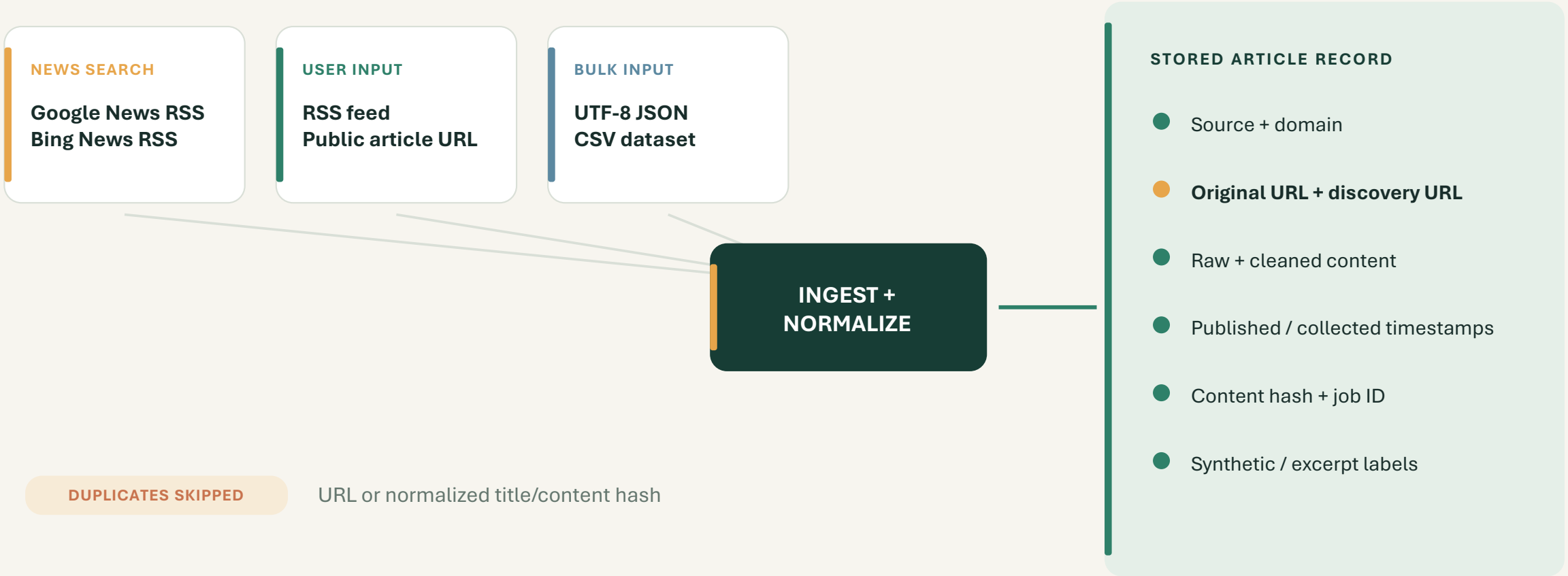
Two runtimes, one evidence-first product model



**JEV activates only when the server-side TypeSafe secret is configured. No key is needed for the local baseline or synthetic demo.*

Collection is a sample with a chain of custody

Each article retains enough metadata to explain where it came from and how it entered the study.



Language is script-detected; relevance is reason-coded

The demo supports English, Hindi, Kannada, Tamil and Telugu, with a configurable Government of India threshold.

EN English

HI Hindi

KN Kannada

TA Tamil

TE Telugu

Language confidence

Script evidence score—not a calibrated probability. Short text, transliteration and code-switching can confuse detection.

Government relevance

Multilingual phrase cues + score + matched reasons. The score supports review; it is not a truth probability.

DEFAULT RELEVANCE THRESHOLD 0.55

Review false matches and missed coverage before drawing conclusions.

Use JEV for bounded decisions—not as a source of evidence

Optional structured classifications for collected, non-synthetic articles; local text evidence remains separate.

OPTIONAL MODEL PROVIDER

TypeSafe AI JEV

Structured choices + probabilities

- GoI relevance
- Sentiment
- Target-specific stance
- Primary topic

Never sends synthetic demo records

01 Key stays server-side

Set TYPESAFE_API_KEY as a Cloudflare Worker secret. Never expose it in browser code.

02 Gate low confidence

Low support maps to uncertain / unclear; stance is used only when a policy target is present.

03 Keep evidence distinct

JEV returns decisions, not quote-level rationale. Local snippets and summaries stay separate.

04 Review the article

Probabilities are model outputs—not verified facts or calibrated certainty.

Sentiment and policy stance answer different questions

OVERALL TONE

FAVORABLE

CRITICAL

POSITIVE TONE / SUPPORTIVE STANCE

Reports benefits and supports the specified policy.

POSITIVE TONE / CRITICAL STANCE

Notes a benefit while questioning design or delivery.

NEGATIVE TONE / SUPPORTIVE STANCE

Supports the goal but reports a setback or delay.

NEGATIVE TONE / CRITICAL STANCE

Criticizes both results and the policy approach.

STANCE TARGET

Union Government policy / action

Store the interpretation next to its support

INPUT

ARTICLE TEXT

Title + cleaned content
Source, date, language
Raw version retained

→

DECISIONS

ANALYSIS

Summary + key points
Sentiment + stance
Topics, entities, claims
Concerns + positive aspects

→

TRACE

EVIDENCE

Supporting text snippets
Confidence + model label
Timestamp + provider note
Source URL + article ID

Review rule Model output is an interpretation. Read the full article and inspect the snippet before using it.

Grouping is useful only when its method is visible

Story comparisons aggregate a selected set of linked article records; clustering quality depends on runtime and sample size.

LOCAL PYTHON DEMO

Multilingual embeddings

FastEmbed vectors → cosine similarity → candidate story clusters

If the model cannot load, the local pipeline logs a lexical fallback.

PROVIDER IS REPLACEABLE

CLOUDFLARE WORKER

Query-scoped candidate set

Online results are collected into a research-job group so their coverage can be compared together.

This is a candidate comparison set, not a semantic embedding cluster. Verify event identity before treating reports as the same story.

METHOD DIFFERENCE IS EXPLICIT

Compare what the stored sample contains

The analysis is descriptive and links every aggregate back to the article IDs behind it.

SAMPLE ARTICLES BY LANGUAGE



COMPARISON OUTPUTS

- **Shared** — Topics and named entities across reports
- **By language** — Sample size, sources, tone and topics
- **By source** — Observed coverage fields for each outlet
- **Differences** — Observed topic emphasis and group counts

Read with sample sizes Language or outlet differences may reflect what was retrieved—not only what was reported.

A student can follow one question all the way to evidence

01 OVERVIEW

Start with the stored sample



02 RESEARCH

Define the query

TOPIC / INITIATIVE

DATE RANGE

LANGUAGE SAMPLE

OPTIONAL OUTLETS

SEARCH + ANALYZE

03 ARTICLE

Inspect the record

Synthetic article

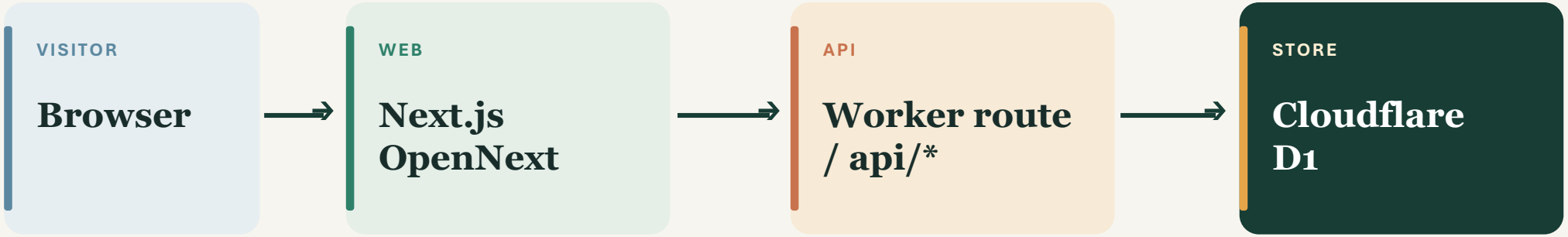
Source, published date, language

Evidence snippet from stored text...

OPEN SOURCE URL

Also reachable via Stories → comparison → linked article

Hosted as a small Workers + D1 prototype



Demo corpus

100 synthetic records preloaded in D1; no paid model credentials required.

Optional live JEV

Set a TypeSafe API key as a Worker secret to enable model decisions on real collected articles.

A custom domain is optional; the default workers.dev host is sufficient for a class demo.

Show the artifacts and checks—not only the diagrams

100

synthetic seed articles in D1

101

stored article analyses

11

candidate story groups incl. live search

5 × 20

language-balanced demo corpus

Acceptance evidence

- API tests cover ingestion, normalization, Unicode, relevance and evidence.
- Live Cloudflare search stored a publisher page with discovery provenance.
- Story and article views expose comparison, evidence and source links.

What is not verified

No human-labeled multilingual benchmark; no validation of JEV with a supplied key; no claim of production-grade security or scale.

CHECK AT PRESENTATION

Do not confuse a useful workflow with a verified conclusion

SAMPLE	Feed discovery is incomplete; publishers may block pages. Excerpts are labeled.
LANGUAGE	Script detection can miss transliteration, code-switching and short text.
MODEL	Tone, stance, relevance and topic labels can be wrong; confidence is not correctness.
CLUSTER	Local embeddings can split or merge stories; the hosted set is query-scoped.
COMPARISON	Different counts may reflect retrieval and sample size, not a real outlet effect.

Never use this prototype alone to claim truth, intent, political bias or overall outlet quality.

Six capability groups define the completed functional slice

- Local foundation + demo dataset
- Ingestion + cleaning + provenance
- Five-language detection + relevance
- Article analysis + evidence
- Story comparison workflow
- API + research UI journey

6 / 10

functional scope

A scope target, not a rubric score.

NEXT

07	Expanded source / language analytics	08	Global search + researcher cluster review	09	Retries, scheduling, pipeline controls	10	Evaluation, release hardening, deployment ops

A clear demo.
A cautious conclusion.

01 Open Overview and point out the synthetic-data label.

02 Search one initiative with aliases, a short date range and two languages.

03 Open the resulting candidate group; compare sample counts and topics.

04 Open an article; show its evidence snippet, model label and source URL.

PROJECT REFERENCES

OpenNext for Cloudflare

opennext.js.org/cloudflare/get-started

Cloudflare Workers

developers.cloudflare.com/workers/

Cloudflare D1

developers.cloudflare.com/d1/

TypeSafe API / JEV

api.typesafe.ai/docs

Project method + limits

docs/METHODOLOGY.md · docs/LIMITATIONS.md

Questions